



DuckDuckGo Mac beta launched

The long-awaited beta launch of DuckDuckGo for Mac is finally live for users to check out.
<https://digit.in/may22-27>



Two thumbs up to you

Netflix has added new "Two Thumbs Up" option for rating content on the streaming platform.
<https://digit.in/may22-28>

Smartphones Batteries and Fast Charging 101

Get started in learning about smartphone batteries and fast charging technology

Dhriti Datta | dhriti@digit.in

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attery and fast charging technology on smartphones is evolving relatively fast. It is no surprise

that battery technology is constantly being developed since it is, after all, one of the most crucial components of your smartphone. However, while most of us know that the batteries in our smartphones power our devices, many don't know the details of how they actually function. Li-ion, mAh, Quick Charge, PD, and other such jargon can get confusing to understand for the uninitiated. So, we're going to delve into the world of smartphone battery and fast charging technology and explain exactly how they work.

HOW BATTERIES IN YOUR SMARTPHONES WORK

The most commonly used type of battery in smartphones is the Lithium-Ion (Li-ion) battery. This is the battery we will be focusing on, however, first, you need to understand how a battery actually functions. Batteries are small vessels or containers of chemical energy. Batteries convert this chemical energy into electrical energy or 'charge'. Inside any typical battery, you have three components – an anode, a cathode and an electrolyte. The anode is the negative end of the battery, often denoted by the '-' sign. The cathode is the positive end of the battery, denoted by '+'.
Source: northwestern.edu



Chemical reactions in the battery cause the build-up of electrons at the anode. An electron is a subatomic particle with a charge of negative energy. The cathode doesn't build up any of the electrons. The electrical difference between the two causes the electrons to flow to the cathode from the anode in an attempt to equalise this difference.

The third element – the electrolyte – acts as an insulator that prevents the electrons from passing through it. The electrolyte is present between the cathode and anode, so the electrons are unable to move directly between them. Now, if you connect the two terminals (anode and cathode) with a wire, a circuit is formed. Electrons can then freely flow through the wire and a current of electricity is produced.

Now, let's get into the specifics of lithium-ion batteries. The concept we explained above stays largely the same in all types of batteries. Smartphone manu-

facturers use lithium-ion batteries since lithium is one of the lightest metals and has great electrochemical potential. Li-ion batteries have a metal oxide cathode, a porous graphite anode and a lithium salt electrolyte. Positively charged lithium ions make their way from the anode to the cathode, thereby driving electrons through the smartphone to power it, and then back again into the anode.

Each smartphone battery has a capacity which

translates to the amount of power your phone's battery can hold. This is technically the amount of lithium ions that are within the anode and cathode. A smartphone's battery capacity is measured in mAh or milliampere-hour. When your phone battery is discharging, the anode releases lithium ions to the cathode. When you plug the phone back into a charger, the opposite happens. Lithium ions are released by the cathode and received by the anode.

One thing to note about lithium-ion batteries is battery lifespan – it is the reason why your smartphones lose their ability to hold a charge for as long as they used to. With each charge

and discharge, the amount of lithium ions within the cathode and anode diminishes. The chemical reaction that produces electricity actually causes some lithium to be trapped and this reduces the number of Li-ions that can move around between the electrodes.

